10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 2.1 — Solving Quadratic Equations

Unit 2 | Solved Exercise | New Book (PECTAA)

Exercise 2.1 (Solved)

1. Write the following quadratic equations in standard form:

(i) $3x - 1 = 2x^2$

(ii) $2x(x + 1) = 4(2x + 3)$

(iii) $2x^2 - 4x = 4x + 7$

(iv) $4(3x - 2) = 9x^2$

(v) $2x + \frac{1}{x} = 5 - \frac{1}{x}$, $x \neq 0$

(vi) $\frac{6x + 6}{20 - x} = \frac{1}{x}$, $x \neq 0, 20$

Standard form of a quadratic equation is $ax^2 + bx + c = 0$, where $a \neq 0$. Bring every term to one side, then arrange the powers of x in descending order.

(i) $3x - 1 = 2x^2$

$2x^2 = 3x - 1$

Answer: $2x^2 - 3x + 1 = 0$

(ii) $2x(x + 1) = 4(2x + 3)$

$2x^2 + 2x = 8x + 12$

$2x^2 - 6x - 12 = 0$

Dividing by 2:

Answer: $x^2 - 3x - 6 = 0$

(iii) $2x^2 - 4x = 4x + 7$

$2x^2 - 4x - 4x - 7 = 0$

Answer: $2x^2 - 8x - 7 = 0$

(iv) $4(3x - 2) = 9x^2$

$12x - 8 = 9x^2$

$9x^2 - 12x + 8 = 0$

Answer: $9x^2 - 12x + 8 = 0$

(v) $2x + 1/x = 5 - 1/x$, $x \neq 0$

Multiply both sides by x:

$2x^2 + 1 = 5x - 1$

$2x^2 - 5x + 2 = 0$

Answer: $2x^2 - 5x + 2 = 0$

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(vi) $(6x + 6)/(20 - x) = 1/x,\ x \neq 0, 20$

Cross multiply:

$x(6x + 6) = 20 - x$

$6x^2 + 6x = 20 - x$

$6x^2 + 7x - 20 = 0$

Answer: $6x^2 + 7x - 20 = 0$

2. Solve the following quadratic equations by factorization method:

(i) $x^2 - x - 6 = 0$

(ii) $x^2 + 3x - 28 = 0$

(iii) $6x^2 + 13x - 5 = 0$

(iv) $x^2 - \frac{3}{2}x = \frac{9}{2}$

(v) $\frac{3x - 8}{x - 2} = \frac{5x - 2}{x + 5},\ x \neq 2, -5$

(vi) $\frac{1}{x - 1} - \frac{1}{x + 3} = \frac{1}{35},\ x \neq 1, -3$

Split the middle term into two parts whose sum is the middle coefficient and whose product is (a)(c). Then take common factors in pairs.

(i) $x^2 - x - 6 = 0$

$x^2 - 3x + 2x - 6 = 0$

$x(x - 3) + 2(x - 3) = 0$

$(x - 3)(x + 2) = 0$

$x - 3 = 0$ or $x + 2 = 0$

$x = 3$ or $x = -2$

Answer: Solution set = {3, −2}

(ii) $x^2 + 3x - 28 = 0$

$x^2 + 7x - 4x - 28 = 0$

$x(x + 7) - 4(x + 7) = 0$

$(x + 7)(x - 4) = 0$

$x + 7 = 0$ or $x - 4 = 0$

Answer: Solution set = {−7, 4}

(iii) $6x^2 + 13x - 5 = 0$

$6x^2 + 15x - 2x - 5 = 0$

$3x(2x + 5) - 1(2x + 5) = 0$

$(2x + 5)(3x - 1) = 0$

$2x + 5 = 0$ or $3x - 1 = 0$

$x = -\frac{5}{2}$ or $x = \frac{1}{3}$

Answer: Solution set = {−5/2, 1/3}

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(iv) $x^2 - (3/2)x = 9/2$

Multiply both sides by 2:

$2x^2 - 3x = 9$

$2x^2 - 3x - 9 = 0$

$2x^2 - 6x + 3x - 9 = 0$

$2x(x - 3) + 3(x - 3) = 0$

$(x - 3)(2x + 3) = 0$

$x = 3 \quad \text{or} \quad x = -\frac{3}{2}$

Answer: Solution set = {3, −3/2}

(v) $(3x - 8)/(x - 2) = (5x - 2)/(x + 5),\ x \neq 2, -5$

Cross multiply:

$(3x - 8)(x + 5) = (5x - 2)(x - 2)$

$3x^2 + 15x - 8x - 40 = 5x^2 - 10x - 2x + 4$

$3x^2 + 7x - 40 = 5x^2 - 12x + 4$

$2x^2 - 19x + 44 = 0$

$2x^2 - 8x - 11x + 44 = 0$

$2x(x - 4) - 11(x - 4) = 0$

$(x - 4)(2x - 11) = 0$

$x = 4 \quad \text{or} \quad x = \frac{11}{2}$

Answer: Solution set = {4, 11/2}

(vi) $1/(x - 1) - 1/(x + 3) = 1/35,\ x \neq 1, -3$

$$\frac{(x + 3) - (x - 1)}{(x - 1)(x + 3)} = \frac{1}{35}$$

$$\frac{4}{x^2 + 2x - 3} = \frac{1}{35}$$

$x^2 + 2x - 3 = 140$

$x^2 + 2x - 143 = 0$

$x^2 + 13x - 11x - 143 = 0$

$x(x + 13) - 11(x + 13) = 0$

$(x + 13)(x - 11) = 0$

$x = -13 \quad \text{or} \quad x = 11$

Answer: Solution set = {−13, 11}

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3. Solve the following quadratic equations by completing square method:

(i) $2x^2 + 5x + 2 = 0$

(ii) $x^2 + x = 42$

(iii) $12x^2 + 7x = 12$

(iv) $\frac{x+3}{2x-7} = \frac{2x-1}{x-3}$, $x \neq \frac{7}{2}, 3$

(v) $\frac{1}{1+x} - \frac{1}{3-x} = \frac{6}{35}$, $x \neq -1, 3$

(vi) $\frac{3x-1}{4x+7} = 1 - \frac{6}{x+4}$, $x \neq -\frac{7}{4}, -4$

Method: make the coefficient of x^2 equal to 1, shift the constant to the right side, add (half the coefficient of $x)^2$ on both sides, and take square root.

(i) $2x^2 + 5x + 2 = 0$

Divide by 2:

$x^2 + \frac{5}{2}x + 1 = 0$

$x^2 + \frac{5}{2}x = -1$

Add $(\frac{5}{4})^2$ on both sides:

$x^2 + \frac{5}{2}x + \frac{25}{16} = -1 + \frac{25}{16}$

$(x + \frac{5}{4})^2 = \frac{9}{16}$

$x + \frac{5}{4} = \pm\frac{3}{4}$

$x = -\frac{5}{4} + \frac{3}{4}$ or $x = -\frac{5}{4} - \frac{3}{4}$

$x = -\frac{1}{2}$ or $x = -2$

Answer: Solution set = {−1/2, −2}

(ii) $x^2 + x = 42$

Add $(\frac{1}{2})^2$ on both sides:

$x^2 + x + \frac{1}{4} = 42 + \frac{1}{4}$

$(x + \frac{1}{2})^2 = \frac{169}{4}$

$x + \frac{1}{2} = \pm\frac{13}{2}$

$x = -\frac{1}{2} + \frac{13}{2}$ or $x = -\frac{1}{2} - \frac{13}{2}$

Answer: Solution set = {6, −7}

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(iii) $12x^2 + 7x = 12$

Divide by 12:

$x^2 + \frac{7}{12}x = 1$

Add $(\frac{7}{24})^2$ on both sides:

$x^2 + \frac{7}{12}x + \frac{49}{576} = 1 + \frac{49}{576}$

$(x + \frac{7}{24})^2 = \frac{625}{576}$

$x + \frac{7}{24} = \pm\frac{25}{24}$

$x = -\frac{7}{24} + \frac{25}{24}$ or $x = -\frac{7}{24} - \frac{25}{24}$

$x = \frac{18}{24}$ or $x = -\frac{32}{24}$

Answer: Solution set = {3/4, −4/3}

(iv) $(x + 3)/(2x - 7) = (2x - 1)/(x - 3)$, $x \neq 7/2, 3$

Cross multiply:

$(x + 3)(x - 3) = (2x - 1)(2x - 7)$

$x^2 - 9 = 4x^2 - 16x + 7$

$3x^2 - 16x + 16 = 0$

Divide by 3:

$x^2 - \frac{16}{3}x + \frac{16}{3} = 0$

$x^2 - \frac{16}{3}x = -\frac{16}{3}$

Add $(\frac{8}{3})^2$ on both sides:

$x^2 - \frac{16}{3}x + \frac{64}{9} = -\frac{16}{3} + \frac{64}{9}$

$(x - \frac{8}{3})^2 = \frac{16}{9}$

$x - \frac{8}{3} = \pm\frac{4}{3}$

$x = \frac{8}{3} + \frac{4}{3}$ or $x = \frac{8}{3} - \frac{4}{3}$

Answer: Solution set = {4, 4/3}

(v) $1/(1 + x) - 1/(3 - x) = 6/35$, $x \neq -1, 3$

$\frac{(3 - x) - (1 + x)}{(1 + x)(3 - x)} = \frac{6}{35}$

$\frac{2 - 2x}{3 + 2x - x^2} = \frac{6}{35}$

$35(2 - 2x) = 6(3 + 2x - x^2)$

$70 - 70x = 18 + 12x - 6x^2$

$6x^2 - 82x + 52 = 0$

Divide by 2:

$3x^2 - 41x + 26 = 0$

$x^2 - \frac{41}{3}x + \frac{26}{3} = 0$

$x^2 - \frac{41}{3}x = -\frac{26}{3}$

Add $(\frac{41}{6})^2$ on both sides:

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$x^2 - \frac{41}{3}x + \frac{1681}{36} = -\frac{26}{3} + \frac{1681}{36}$

$(x - \frac{41}{6})^2 = \frac{1369}{36}$

$x - \frac{41}{6} = \pm\frac{37}{6}$

$x = \frac{41}{6} + \frac{37}{6}$ or $x = \frac{41}{6} - \frac{37}{6}$

$x = 13$ or $x = \frac{2}{3}$

Answer: Solution set = {13, 2/3}

(vi) $(3x - 1)/(4x + 7) = 1 - 6/(x + 4)$, $x \neq -7/4, -4$

$1 - \frac{6}{x+4} = \frac{x+4-6}{x+4}$

$= \frac{x-2}{x+4}$

$\frac{3x-1}{4x+7} = \frac{x-2}{x+4}$

Cross multiply:

$(3x - 1)(x + 4) = (x - 2)(4x + 7)$

$3x^2 + 12x - x - 4 = 4x^2 + 7x - 8x - 14$

$3x^2 + 11x - 4 = 4x^2 - x - 14$

$x^2 - 12x - 10 = 0$

$x^2 - 12x = 10$

Add $(6)^2$ on both sides:

$x^2 - 12x + 36 = 10 + 36$

$(x - 6)^2 = 46$

$x - 6 = \pm\sqrt{46}$

Answer: $x = 6 \pm \sqrt{46}$, solution set = {6 + √46, 6 − √46}

4. Use quadratic formula to solve the following equations:

(i) $2x^2 - 5x + 3 = 0$

(ii) $2x^2 - 7x - 15 = 0$

(iii) $2x^2 + 7x = 15$

(iv) $x^2 + 11 = 7x$

(v) $\frac{x+4}{x-4} + \frac{x-2}{x-3} = 6\frac{1}{3}$, $x \neq$ 4, 3

(vi) $\frac{3x-3}{x+1} = \frac{2x-1}{x-1}$, $x \neq -1, 1$

Quadratic formula: $x = -b \pm \sqrt{(b^2 - 4ac)}/2a$ for $ax^2 + bx + c = 0$.

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(i) $2x^2 - 5x + 3 = 0$

$a = 2,\ b$

$= -5,\ c$

$= 3$

$$x = \frac{-(-5) \pm \sqrt{((-5)^2 - 4(2)(3))}}{2(2)}$$

$$x = \frac{5 \pm \sqrt{(25 - 24)}}{4}$$

$$x = \frac{5 \pm 1}{4}$$

$$x = \frac{6}{4} \quad \text{or} \quad x = \frac{4}{4}$$

Answer: Solution set = {3/2, 1}

(ii) $2x^2 - 7x - 15 = 0$

$a = 2,\ b$

$= -7,\ c$

$= -15$

$$x = \frac{7 \pm \sqrt{((-7)^2 - 4(2)(-15))}}{2(2)}$$

$$x = \frac{7 \pm \sqrt{(49 + 120)}}{4}$$

$$x = \frac{7 \pm \sqrt{169}}{4}$$

$$x = \frac{7 \pm 13}{4}$$

$$x = \frac{20}{4} \quad \text{or} \quad x = \frac{-6}{4}$$

Answer: Solution set = {5, −3/2}

(iii) $2x^2 + 7x = 15$

$2x^2 + 7x - 15 = 0$

$a = 2,\ b$

$= 7,\ c$

$= -15$

$$x = \frac{-7 \pm \sqrt{((7)^2 - 4(2)(-15))}}{2(2)}$$

$$x = \frac{-7 \pm \sqrt{(49 + 120)}}{4}$$

$$x = \frac{-7 \pm 13}{4}$$

$$x = \frac{6}{4} \quad \text{or} \quad x = \frac{-20}{4}$$

Answer: Solution set = {3/2, −5}

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(iv) $x^2 + 11 = 7x$

$x^2 - 7x + 11 = 0$

$a = 1,\ b$

$= -7,\ c$

$= 11$

$$x = \frac{7 \pm \sqrt{((-7)^2 - 4(1)(11))}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{(49 - 44)}}{2}$$

Answer: x = (7 ± √5)/2, solution set = {(7 + √5)/2, (7 − √5)/2}

(v) $(x + 4)/(x - 4) + (x - 2)/(x - 3) = 6⅓,\ x \neq 4, 3$

$$\frac{(x + 4)(x - 3) + (x - 2)(x - 4)}{(x - 4)(x - 3)} = \frac{19}{3}$$

$$\frac{x^2 + x - 12 + x^2 - 6x + 8}{x^2 - 7x + 12} = \frac{19}{3}$$

$$\frac{2x^2 - 5x - 4}{x^2 - 7x + 12} = \frac{19}{3}$$

$3(2x^2 - 5x - 4) = 19(x^2 - 7x + 12)$

$6x^2 - 15x - 12 = 19x^2 - 133x + 228$

$13x^2 - 118x + 240 = 0$

$a = 13,\ b$

$= -118,\ c$

$= 240$

$$x = \frac{118 \pm \sqrt{((-118)^2 - 4(13)(240))}}{2(13)}$$

$$x = \frac{118 \pm \sqrt{(13924 - 12480)}}{26}$$

$$x = \frac{118 \pm \sqrt{1444}}{26}$$

$$x = \frac{118 \pm 38}{26}$$

$$x = \frac{156}{26} \quad \text{or} \quad x = \frac{80}{26}$$

Answer: Solution set = {6, 40/13}

(vi) $(3x - 3)/(x + 1) = (2x - 1)/(x - 1),\ x \neq -1, 1$

Cross multiply:

$(3x - 3)(x - 1) = (2x - 1)(x + 1)$

$3x^2 - 3x - 3x + 3 = 2x^2 + 2x - x - 1$

$3x^2 - 6x + 3 = 2x^2 + x - 1$

$x^2 - 7x + 4 = 0$

$a = 1,\ b$

$= -7,\ c$

$= 4$

$$x = \frac{7 \pm \sqrt{((-7)^2 - 4(1)(4))}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{(49 - 16)}}{2}$$

Answer: x = (7 ± √33)/2, solution set = {(7 + √33)/2, (7 − √33)/2}

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5. Solve the following quadratic equations graphically:

(i) $x^2 - 3x - 18 = 0$ (ii) $x^2 - 5x - 14 = 0$

(iii) $2x^2 + 13x + 6 = 0$ (iv) $4x^2 + 12x - 27 = 0$

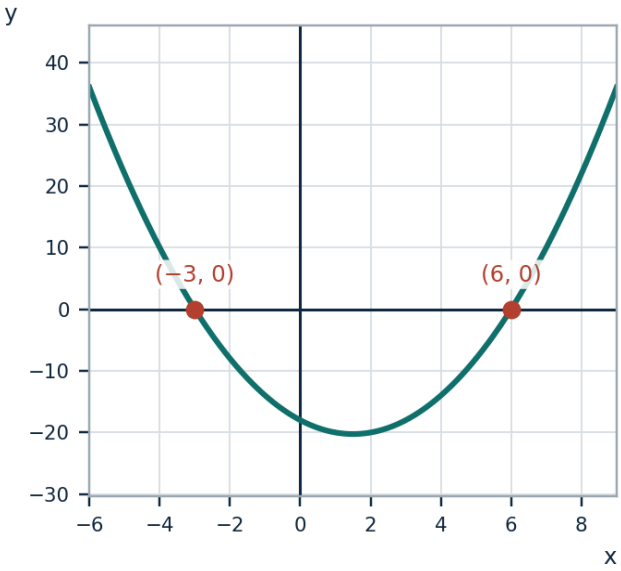
Method: write $y = ax^2 + bx + c$, make a table of values, plot the points and join them by a smooth curve. The x-coordinates of the points where the curve cuts the x-axis are the roots.

(i) $x^2 - 3x - 18 = 0$

Let $y = x^2 - 3x - 18$

Table of values

x	−2	−1	1	2	5	
y	−8	12	−26	−20	−4	2



The curve cuts the x-axis at $x = -3$ and $x = 6$

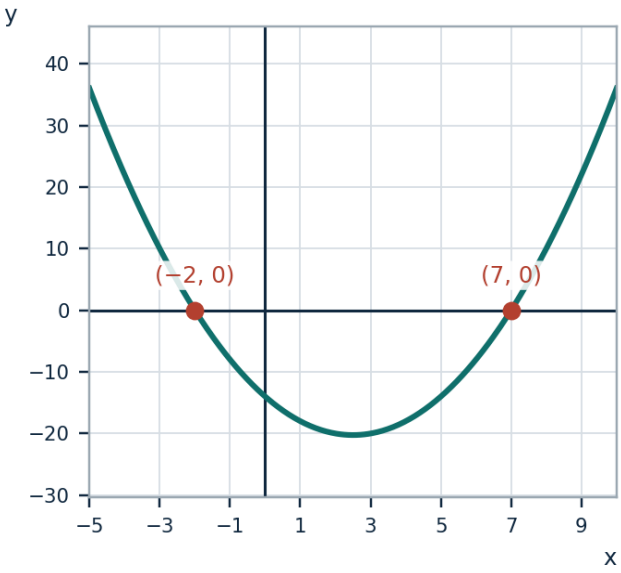
Answer: Solution set = {−3, 6}

(ii) $x^2 - 5x - 14 = 0$

Let $y = x^2 - 5x - 14$

Table of values

x	−1	0	2	3	6
y	8	−8	−20	−20	4



The curve cuts the x-axis at x = −2 and x = 7

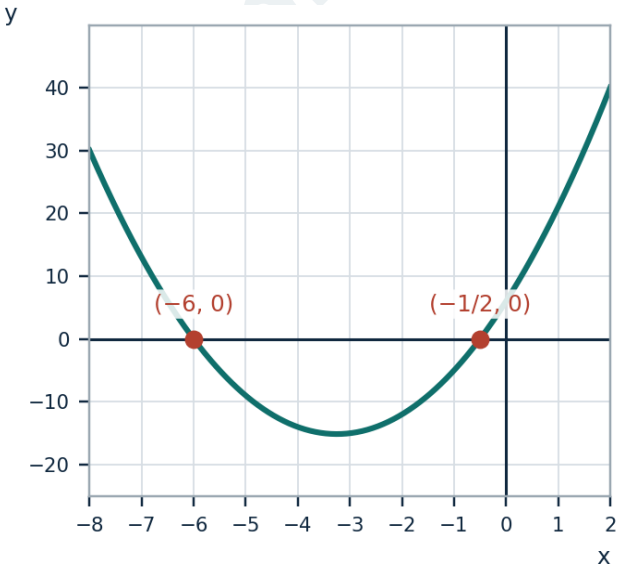
Answer: Solution set = {−2, 7}

(iii) $2x^2 + 13x + 6 = 0$

Let $y = 2x^2 + 13x + 6$

Table of values

x	−7	−5	−3	−1	1
y	8	−14	−16	−5	21



The curve cuts the x-axis at x = −6 and x = −1/2

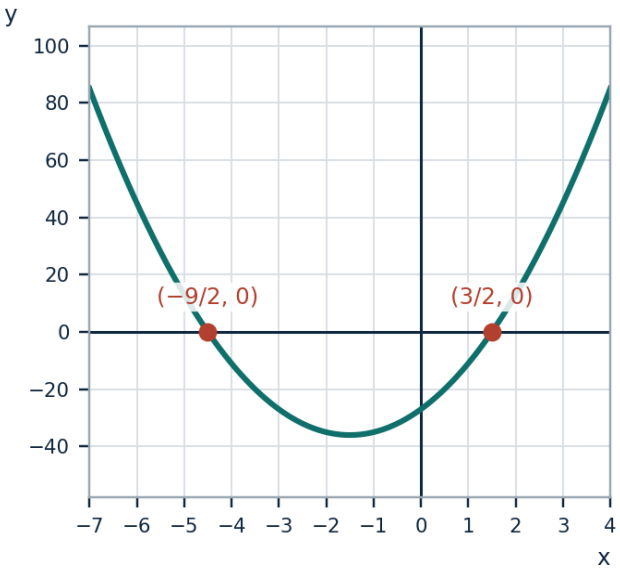
Answer: Solution set = {−6, −1/2}

(iv) $4x^2 + 12x - 27 = 0$

Let $y = 4x^2 + 12x - 27$

Table of values

x	−5	−3	−1	1	2
y	13	−27	−35	−11	5



The curve cuts the x-axis at x = −9/2 and x = 3/2

Answer: Solution set = {−9/2, 3/2}

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